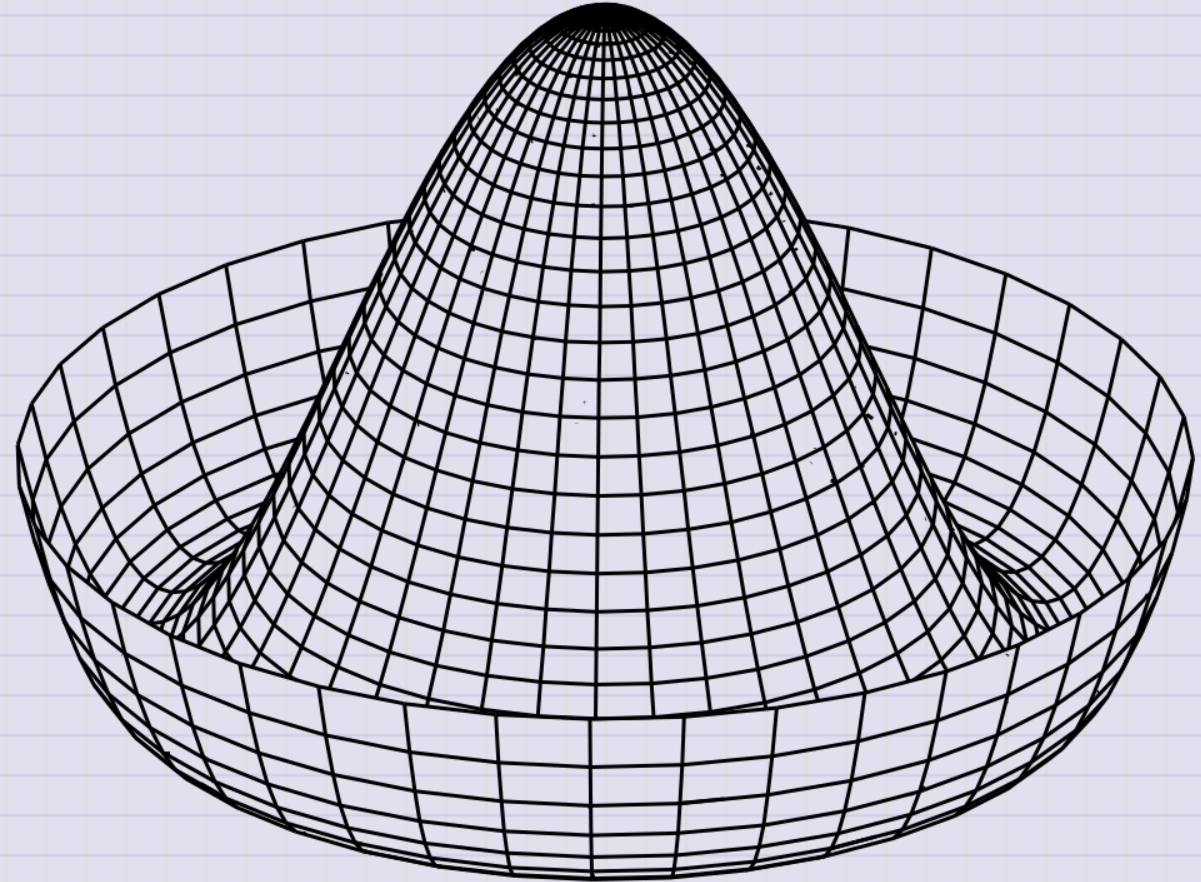


Day 15 - Potential Energy and Stability

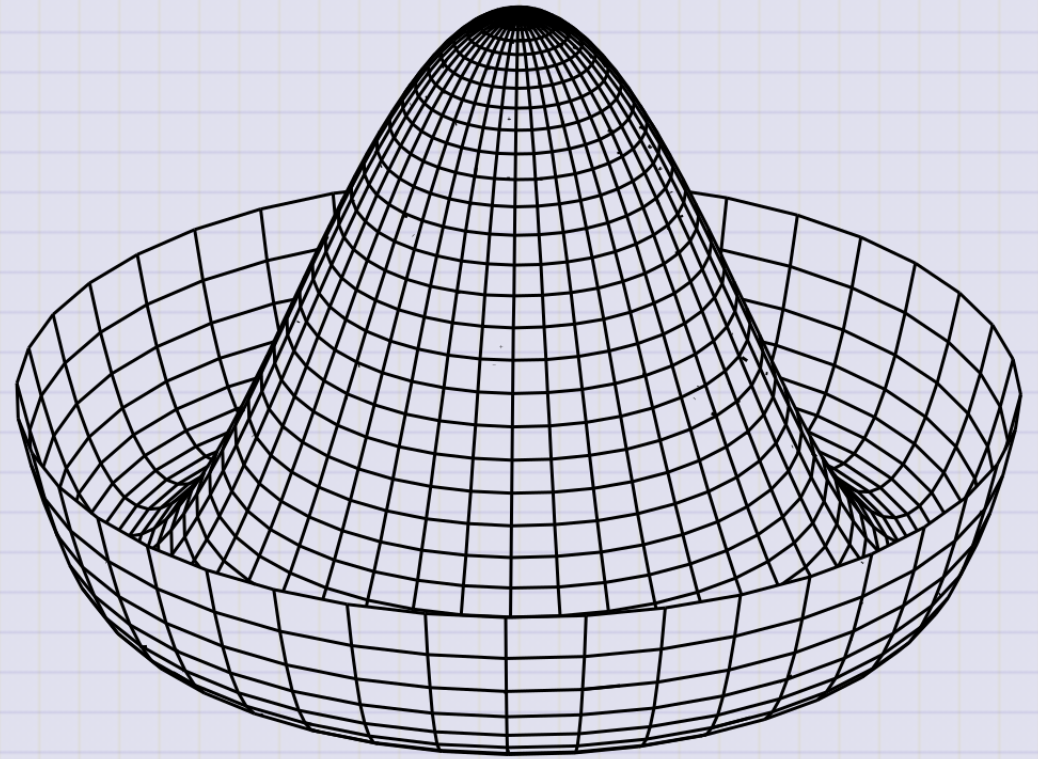
Mexican Hat/Sombrero Potential $\longrightarrow$



Mexican Hat Potential

$$V(\phi) = -5|\phi|^2 + |\phi|^4$$

- Spontaneous Symmetry Breaking (Jeffery Goldstone, 1961)
- Unstable vacuum state at $\phi = 0$
 - Peak of the hat
- Infinite number of stable minima
 - $\phi = \sqrt{5/2}e^{i\phi}$



Announcements

- HW 4 is due today
- Midterm 1 is available today (Due Feb 24th)

Seminars this week

This Week's Goals

- Understand the concept of potential energy
- Determine the equilibrium points of a system using potential energy
- Analyze the stability of equilibrium points
- Define and begin to apply conservation of linear and angular momentum

Reminders: Conservative Forces

- Conservative forces are those with zero curl

$$\nabla \times \vec{F} = 0$$

- The work done by a conservative force is path-independent; on a closed path, the work done is zero

$$\oint \vec{F} \cdot d\vec{r} = 0$$

- The force can be written as the gradient of a scalar potential energy function

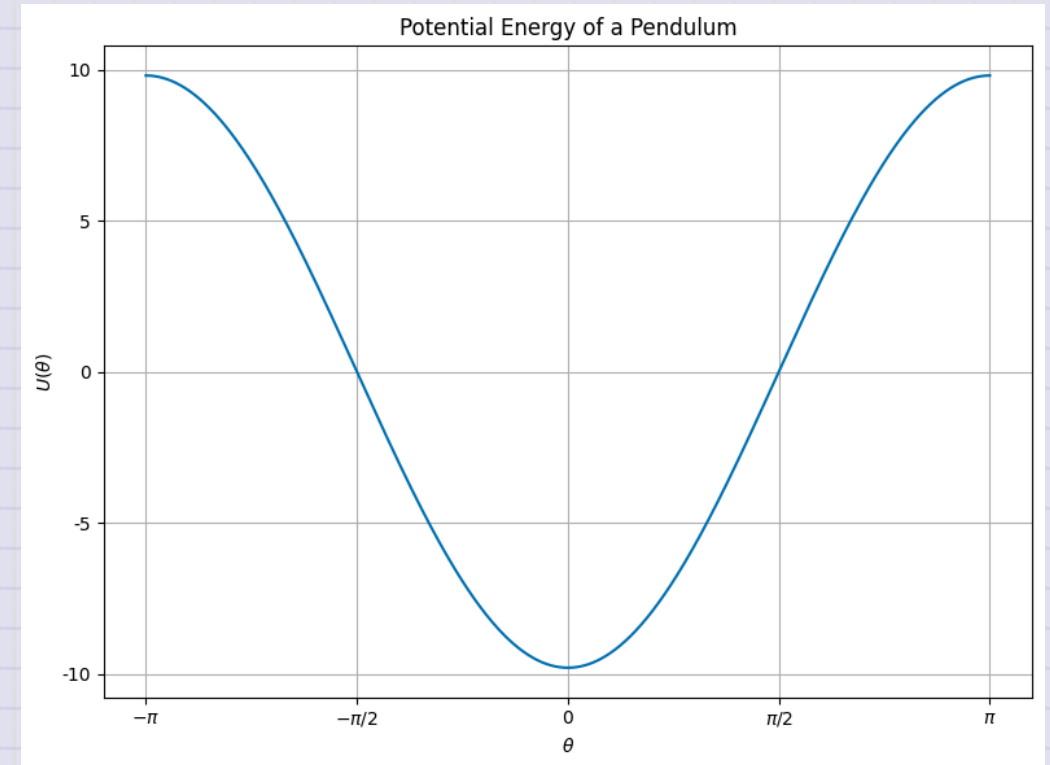
$$\vec{F} = -\nabla U$$

Clicker Question 15-1

Here's the graph of the potential energy function $U(x)$ for a pendulum.

Using this graph, what can you say about the equilibrium points of the pendulum? There is/are:

1. One stable point
2. Two stable points
3. One stable and one unstable point
4. Two unstable and one stable point



Clicker Question 15-2

Here's a potential energy function $U(x)$ for a pendulum:

$$U(\phi) = -mgL \cos(\phi) + U_0$$

1. Find the equilibrium points (ϕ^*) of the pendulum by setting:

$$\frac{dU(\phi^*)}{d\phi} = 0.$$

2. Characterize the stability of the equilibrium points (ϕ^*) by examining the second derivative:

$$\frac{d^2U(\phi^*)}{d\phi^2} > 0? \quad \frac{d^2U(\phi^*)}{d\phi^2} < 0?$$

Click when done.

Clicker Question 15-3

A double-well potential energy function $U(x)$ is given by

$$U(x) = -\frac{1}{2}kx^2 + \frac{1}{4}kx^4.$$

We assume we have scaled the potential energy so that all the units are consistent.

How many equilibrium points does this system have?

- 1. 1
- 2. 2
- 3. 3
- 4. 4

Clicker Question 15-4

A double-well potential energy function $U(x)$ is given by

$$U(x) = -\frac{1}{2}kx^2 + \frac{1}{4}kx^4.$$

1. Find the equilibrium points (x^*) of the pendulum by setting:

$$\frac{dU(x^*)}{dx} = 0.$$

2. Characterize the stability of the equilibrium points (x^*) by examining the second derivative:

$$\frac{d^2U(x^*)}{dx^2} > 0? \quad \frac{d^2U(x^*)}{dx^2} < 0?$$

Click when done.

Clicker Question 15-5

Here's a graph of the potential energy function $U(x)$ for a double-well potential.

Describe the motion of a particle with the total energy, $E =$

1. 0.4 J , $<$ barrier height
2. 1.2 J , $>$ barrier height
3. 1.0 J , $=$ barrier height

Click when done.

